30 · Semen, sperm count, erection and ejaculation

The words of this track, as it says them. Bold brown is the examiner.

Semen, sperm count, erection and ejaculation.

Describe the composition of semen, and the normal sperm count in relation to sterility. Then describe the physiology of the male sexual response, and relate it to erectile dysfunction.

Two checklist points. Her quizzes ask the semen part as match-the-gland, and the erection part as put-the-steps-in-order.

The hook for semen: sixty, thirty, ten. Sixty per cent from the seminal vesicles, thirty from the prostate, and ten from everything else: the sperm themselves, the testes, the epididymis, and the bulbourethral glands.

Semen is a milky, sticky mixture of sperm and the accessory gland fluids. It gives the sperm a transport medium; nutrients, fructose from the seminal vesicles and citrate from the prostate; protection; and it activates them. It's alkaline, which neutralises the acid in the male urethra and in the vagina. Its prostaglandins thin the mucus in the cervix, and set off reverse peristalsis in the uterus, which helps the sperm on their way. Clotting factors make it sticky straight after ejaculation, then an enzyme, fibrinolysin, liquefies it again.

The count. Only two to five millilitres is ejaculated, but it holds fifty to a hundred and thirty million sperm in every millilitre. Her quiz key: as many as seven hundred and fifty million sperm can be released in one ejaculation. Sterility is linked to a low count: below about twenty to twenty-five million per millilitre.

Now the male sexual response. The hook: point and shoot. Parasympathetic for the erection, sympathetic for the ejaculation.

Erection. Sexual arousal sets off a parasympathetic reflex, which releases nitric oxide locally. Nitric oxide relaxes the smooth muscle in the vessel walls, so the arterioles dilate and the erectile tissue fills with blood. The corpora cavernosa expand, and that squashes the veins that drain them, so the blood stays in. The corpus spongiosum keeps the urethra open. An erection can be started, or stopped, by thoughts and emotions alone.

Ejaculation. Sympathetic nerves make the ducts and the accessory glands contract and empty into the urethra. The bladder sphincter closes, so no urine gets out. And the bulbospongiosus muscles contract rhythmically, pumping the semen out of the urethra.

Erectile dysfunction, or impotence, is the inability to attain an erection. Her slide adds how Viagra works: an enzyme called P D E 5 breaks down c G M P, the messenger that keeps the smooth muscle relaxed, and Viagra blocks P D E 5. So the vessels stay open.

Once more. Semen, and the male sexual response.

Semen is sixty per cent from the seminal vesicles, thirty from the prostate, and ten from the sperm and the rest. It carries, feeds, protects and activates the sperm, and it's alkaline. Two to five millilitres holds fifty to a hundred and thirty million sperm per millilitre, and a count below about twenty million is linked to sterility. Erection is parasympathetic: nitric oxide dilates the arterioles, the corpora cavernosa fill, and the veins are squashed shut. Ejaculation is sympathetic: the ducts contract, the bladder sphincter closes, and the bulbospongiosus pumps the semen out. Erectile dysfunction is being unable to attain an erection.